

Network Layer

- Network Layer Characteristics
- IPv4 addressing
- IPv6 addressing
- Sub netting
- Routing Protocols

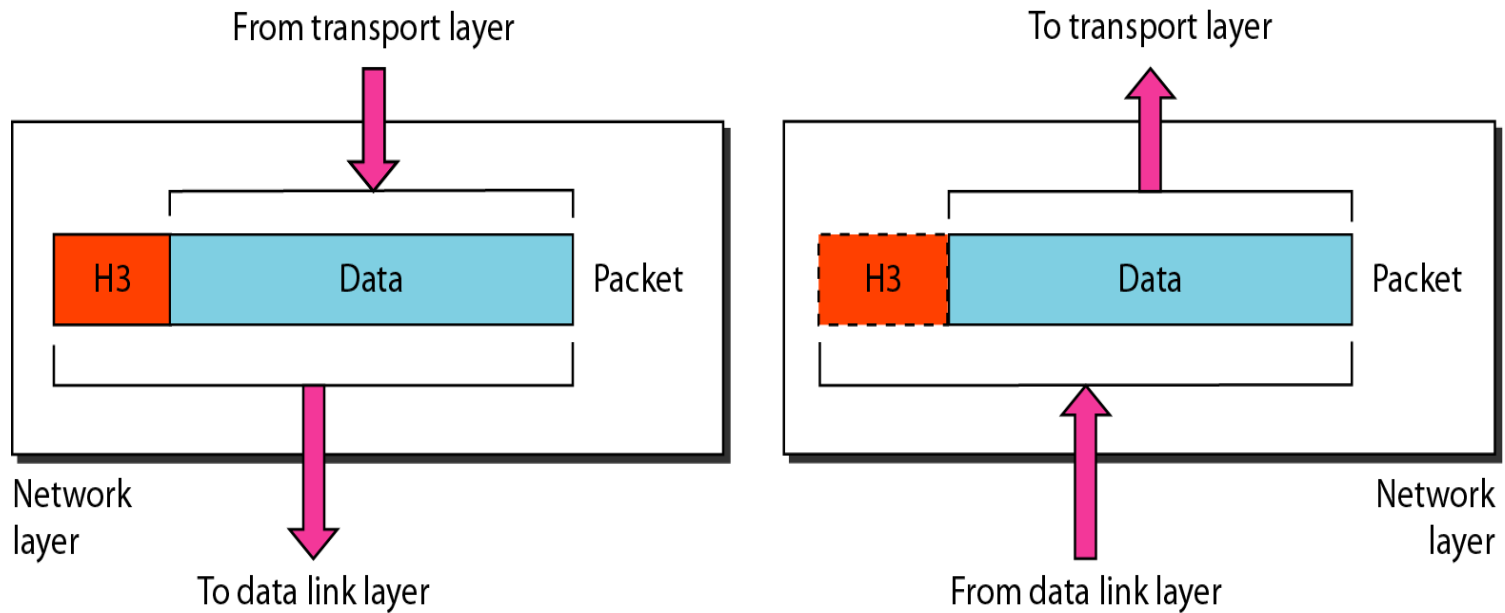
Network Layer (3rd OSI Layer)

- Concerned with getting packets from source to destination.
- The network layer must know the topology of the subnet and choose appropriate paths through it.
- It is responsible for the source-to-destination delivery of a packet, possibly across multiple networks.

Network Layer (3rd OSI Layer)

- the data link layer oversees the delivery of the packet between two systems on the same network, the network layer ensures that each packet gets from its point of origin to its final destination.
- If two systems are connected to the same local network, there is usually no need for a network layer.

Contd.



Network Layer Functionalities

1. Logical addressing

- The **physical addressing** implemented by the data link layer handles the addressing problem **locally**.
- If a packet passes the network boundary, we need another addressing system to help distinguish the source and destination systems.
- The network layer **adds a header to the packet** coming from the upper layer that, among other things, includes the logical addresses of the sender and receiver.

Contd.

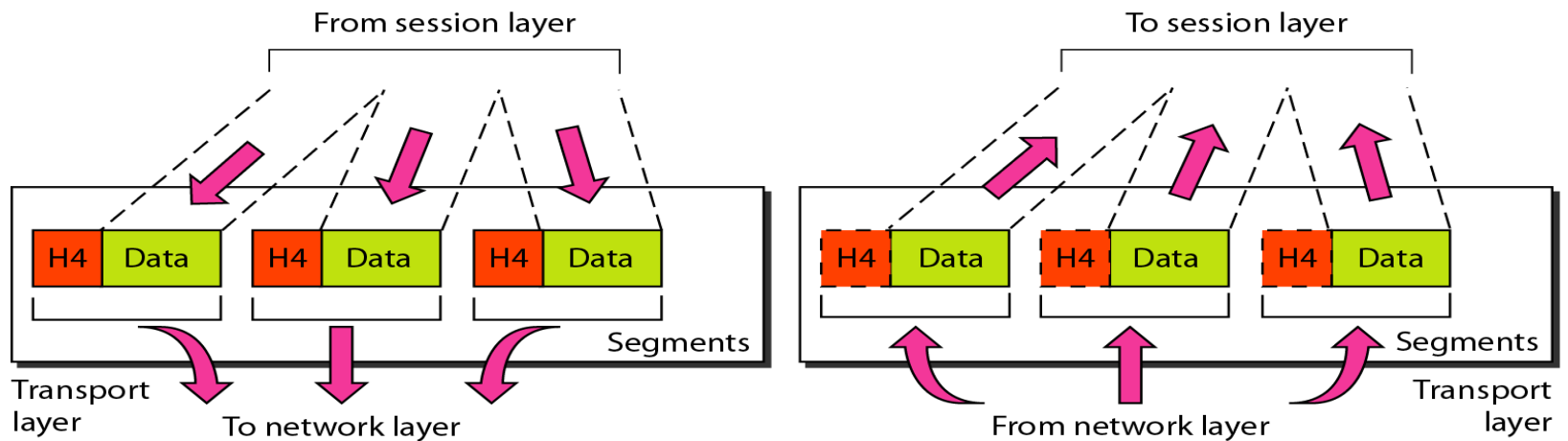
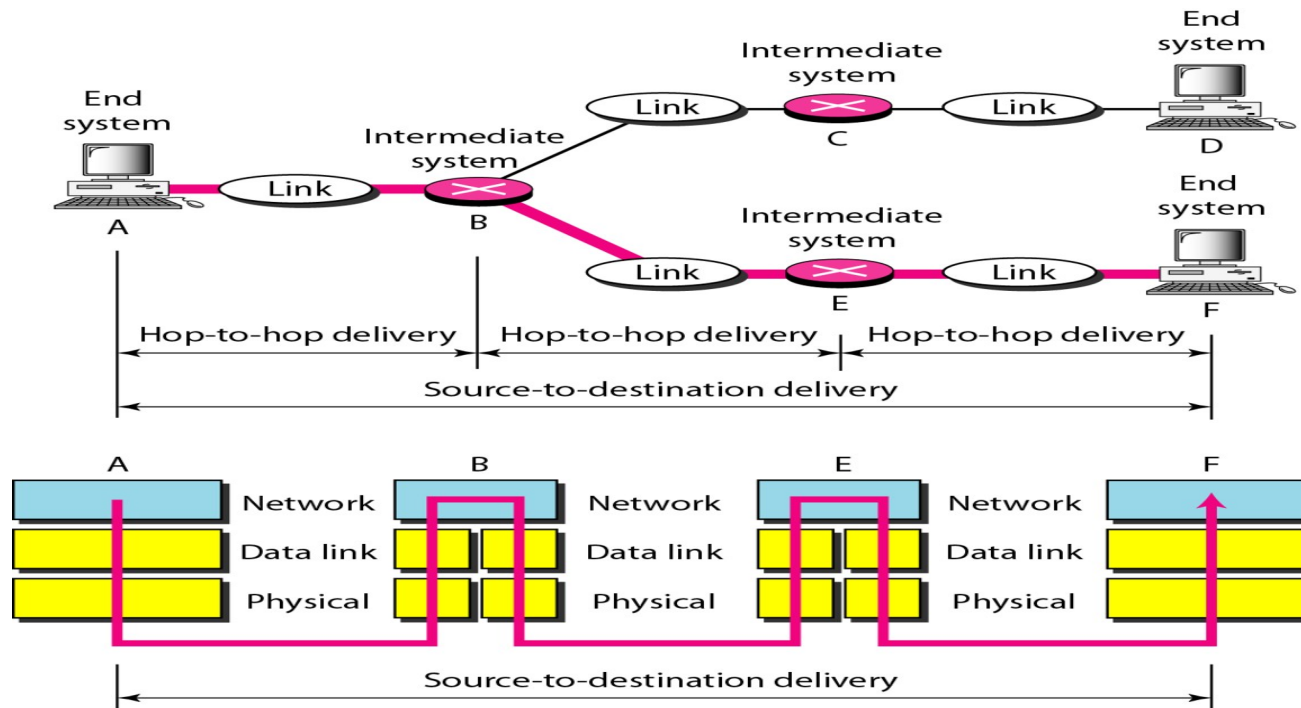
2. Routing

- When independent networks or links are connected to create internetworks (network of networks) or a large network, the connecting devices (called routers or switches) route or switch the packets to their final destination.
- One of the functions of the network layer is to provide this mechanism.

Contd.

3. Internetworking

- Internetworks are multiple networks that are connected in such a way that they act as one large network, connecting multiple office or department networks.
- Internetworks are connected by networking hardware such as routers, switches, and bridges.
- Internetworking is a solution born of three networking problems:
 - » Isolated LANs
 - » duplication of resources
 - » the lack of a centralized network management system.



Internet Protocol (IP)

- used for communicating data across a packet-switched internetwork using the IP Suite, also referred to as TCP/IP.
- IP is the primary protocol in the Internet Layer of the Internet Protocol Suite
- IP has the task of delivering distinguished protocol datagrams (packets) from the source host to the destination host solely based on their addresses.

Internet Protocol (IP)

- IP defines addressing methods and structures for datagram encapsulation.
- The first major version of addressing structure, now referred to as Internet Protocol Version 4 (**IPv4**) is still the dominant protocol of the Internet
- although the successor, Internet Protocol Version 6 (**IPv6**), is being deployed actively worldwide (128 bits).

Contd.

- Communication at the network layer is **host-to-host** (**computer-to-computer**);
 - a computer somewhere in the world needs to communicate with another computer somewhere else in the world. Usually, computers communicate through the Internet.

Contd.

- The packet transmitted by the sending computer may pass through several LANs or WANs before reaching the destination computer.
- For this level of communication, we need a global addressing scheme; we use the term IP address to mean a logical address in the network layer of the TCP/IP protocol suite.

IPv4 Addresses

- An IPv4 address is a 32-bit address
- IPv4 addresses are unique and universal.
- They are unique in the sense that each address defines one, and only one, connection to the Internet.
- Two devices on the Internet can never have the same address at the same time.

Address Space

- An address space is the total number of addresses used by the protocol.
- If a protocol uses N bits to define an address, the address space is 2^N because each bit can have two different values (0 or 1) and N bits can have 2^N values.

Address Space

- IPv4 uses 32-bit addresses, i.e. the address space is 2^{32} or 4,294,967,296 (more than 4 billion).
- if there were no restrictions, more than 4 billion devices could be connected to the Internet.
- the actual number is much less because of the restrictions imposed on the addresses.

Notations

- There are two prevalent notations to show an IPv4 address: **binary notation** and **dotted-decimal notation**.
- **Binary Notation**
 - In binary notation, the IPv4 address is displayed as 32 bits.
 - Each octet is often referred to as a byte.
 - a 4-byte address.
 - an example of an IPv4 address in binary notation:

01110101 10010101 00011101 00000010

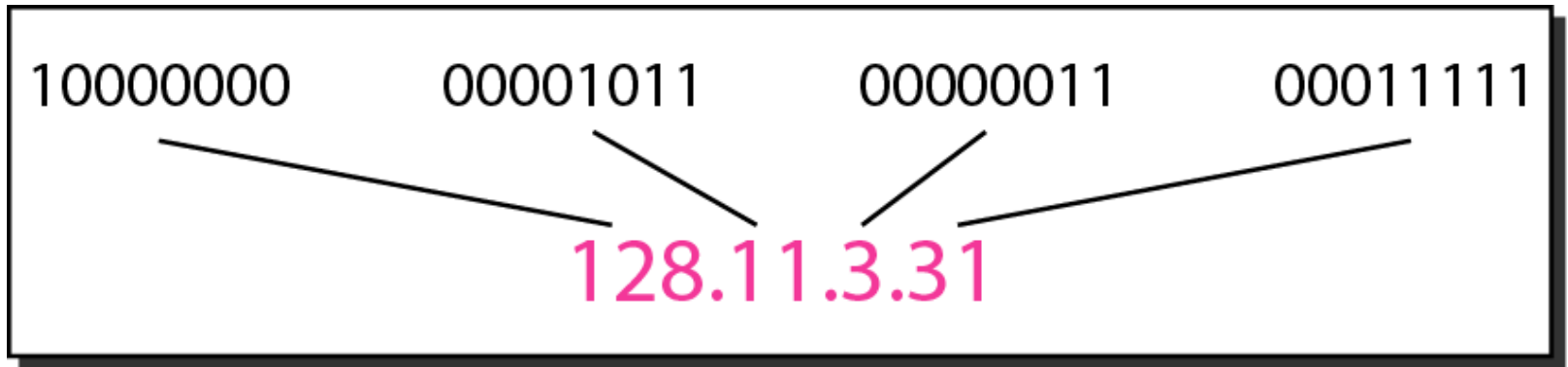
Notations

- **Dotted-Decimal Notation**

- To make the IPv4 address more compact and easier to read, Internet addresses are usually written in decimal form with a decimal point (dot) separating the bytes.
- The following is the dotted-decimal notation of the above address:

117.149.29.2

Notations: Example



Example 1

- Change the following IPv4 addresses from binary notation to dotted-decimal notation

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

- replace each group of 8 bits with its equivalent decimal number and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255

Example 2

- Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

Solution

- Replace each decimal number with its binary equivalent.

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010

Example 3

- Find the error, if any, in the following IPv4 addresses.

c. 75.45.301.14

d. 11100010.23.14.67

Solution

- There must be no leading zero (045).
- There can be no more than four numbers.
- Each number needs to be less than or equal to 255.
- A mixture of binary notation and dotted-decimal notation is not allowed.

Class full and Classless addressing

- **Class full addressing** is an IP address allocation method that allocates IP addresses according to five major classes(A-E).
- **Classless addressing** is an IP address allocation method that is designed to replace classful addressing to minimize the rapid exhaustion of IP addresses.

Class full Addressing

- IPv4 addressing, at its inception, used the concept of **classes**.
- This architecture is called **class full addressing**.
- In class full addressing, the address space is divided into five classes: **A, B, C, D, and E**.
- Each class occupies some part of the address space.

Classful Addressing

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation

Class	HOB	NET ID Bits	Host ID Bits	No of Networks	Host Per Network	Start Address	End Address
Class A	0	8	24	$2^7=128$	$2^{24}=16,777,216$	0.0.0.0	127.255.255.255
Class B	10	16	16	$2^{14}=16,384$	$2^{16}=65,536$	128.0.0.0	191.255.255.255
Class C	110	24	8	$2^{21}=2,097,152$	$2^8=256$	192.0.0.0	223.255.255.255
Class D	1110	-	-	-	-	224.0.0.0	239.255.255.255
Class E	1111	-	-	-	-	240.0.0.0	255.255.255.255

Public IP address	Private IP address
External (global) reach	Internal (local) reach
Used for communicating outside your private network, over the internet	Used for communicating within your private network, with other devices in your home or office
A unique numeric code never reused by other devices	A non-unique numeric code that may be reused by other devices in other private networks
Assigned and controlled by your internet service provider	Assigned to your specific device within a private network
Not free	Free
Any number not included in the reserved private IP address range Example: 8.8.8.8.	10.0.0.0 — 10.255.255.255; 172.16.0.0 — 172.31.255.255; 192.168.0.0 — 192.168.255.255 Example: 10.11.12.13

Example 4

- Find the class of each address.
 - a. 00000001 00001011 00001011 11101111
 - b. 11000001 10000011 00011011 11111111
 - c. 14.23.120.8
 - d. 252.5.15.111

Solution

- a. The first bit is 0. This is a class A address.
- b. The first 2 bits are 1; the third bit is 0. This is a class C address.
- c. The first byte is 14; the class is A.
- d. The first byte is 252; the class is E.

Classes and Blocks

- One problem with class full addressing is that each class is divided into a fixed number of blocks with each block having a fixed size

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Number of Blocks for class A = 2^7

Number of Blocks for class B = 2^{14}

Number of Blocks for class C = 2^{21}

Block size for class A = 2^{24}

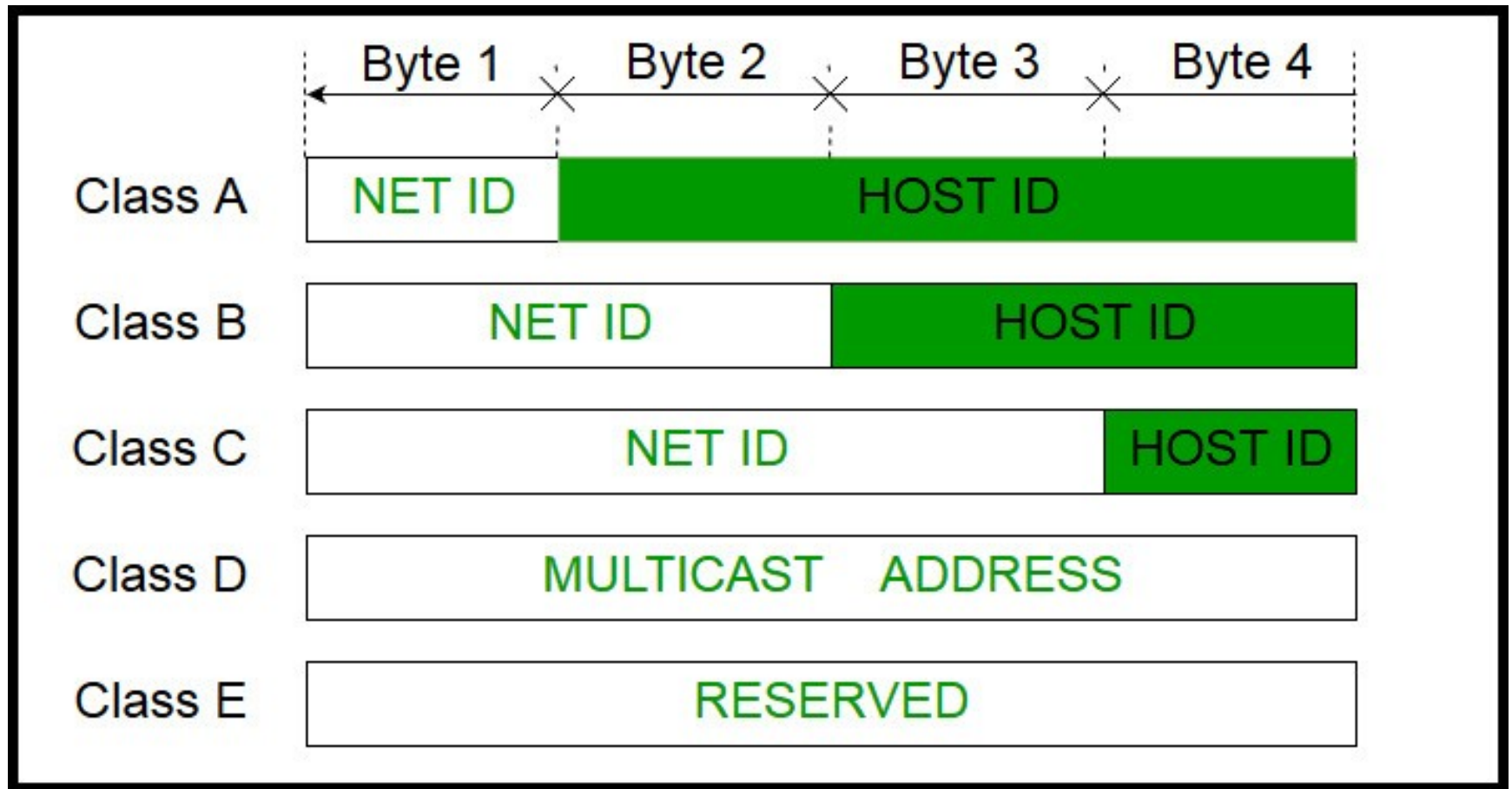
Block size for class B = 2^{16}

Block size for class C = 2^8

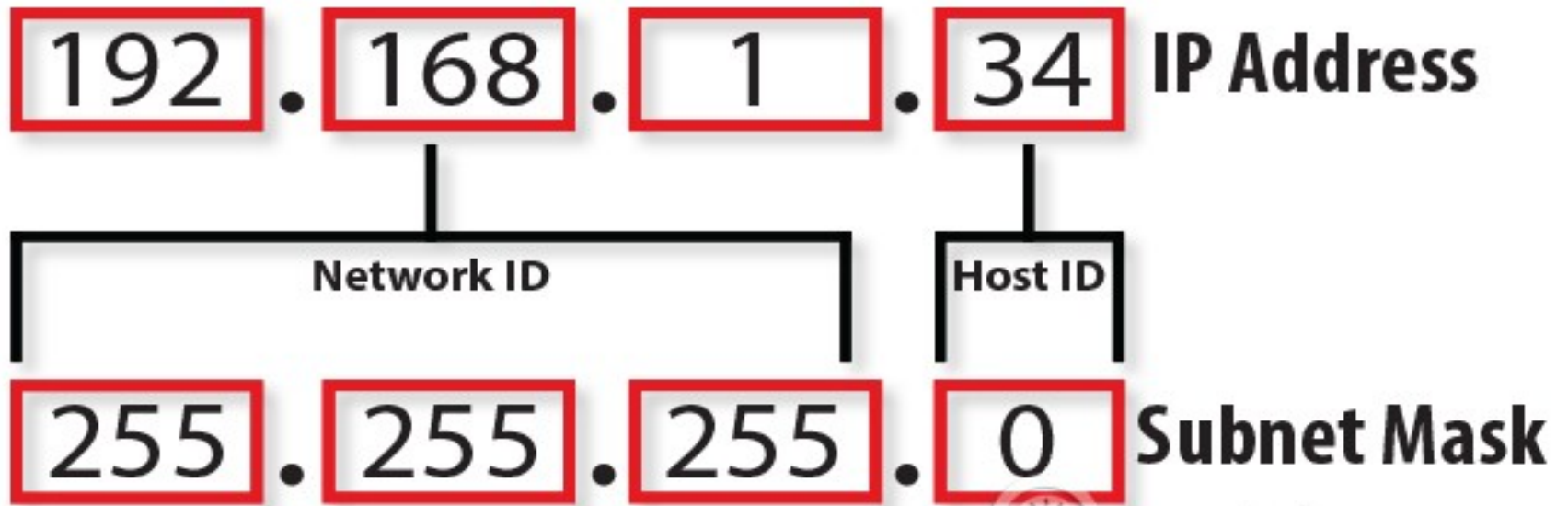
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- **0** [Zero] is reserved and represents all IP addresses.
- **127** is a reserved address and is used for testing, like a loop back on an interface.
- **255** is a reserved address and is used for broadcasting purposes.

Network ID and Host ID



Mask



Mask

- The mask can help us to find the network ID and the host ID.

Class A
Subnet Mask

Network	Host	Host	Host
255	0	0	0

Class B
Subnet Mask

Network	Network	Host	Host
255	255	0	0

Class C
Subnet Mask

Network	Network	Network	Host
255	255	255	0

Mask

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255 .0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255 .0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255 .0	/24

Address Depletion

- The **flaws in class full** addressing scheme combined with the **fast growth of the Internet** led to the **near depletion** of the available addresses.
- Yet the number of devices on the Internet is much **less than the 2^{32}** address space. We have **run out of class A and B addresses**, and a **class C block is too small** for most midsize organizations.
- One solution that has **alleviated** the problem is the idea of **classless addressing**.

Classless Addressing

- To overcome address depletion and give more organizations access to the Internet, **classless addressing** was designed and implemented.
- In this scheme, there are **no classes**, but the addresses are **still granted in blocks**.

Contd.

- A block in **class A address is too large** for almost any organization. This means most of the addresses in class A were wasted and were not used.
- A block in **class B is also very large**, probably too large for many of the organizations that received a class B block.
- A block in **class C is probably too small** for many organizations.

Classless Addressing

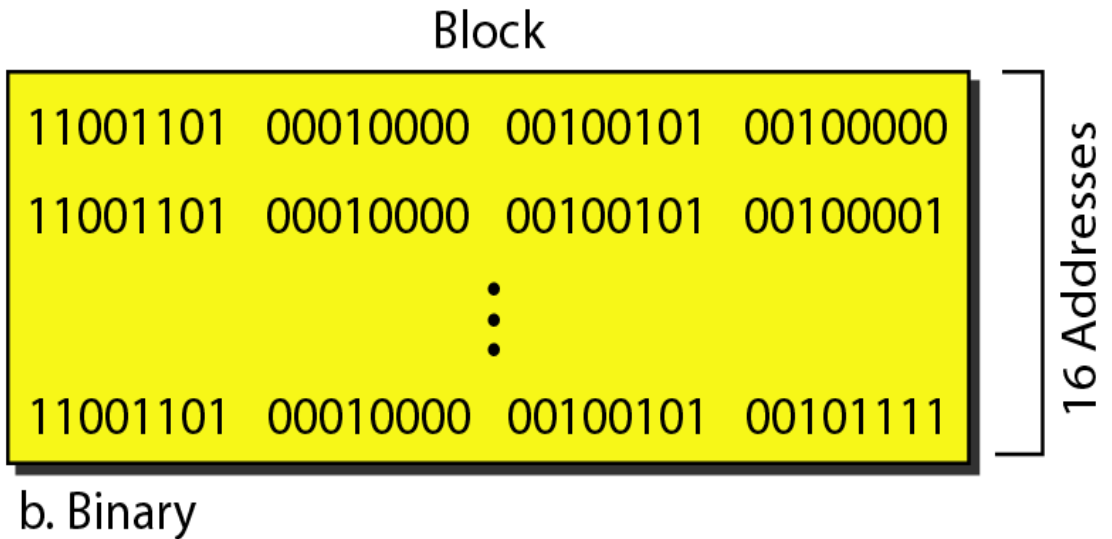
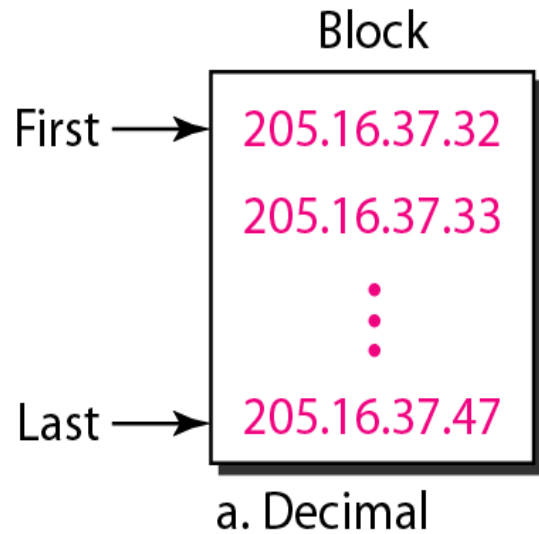
■ Address Blocks

- The size of the block (**the number of addresses**) varies based on the **nature and size** of the entity.
- For example,
 - a **household** may be given only **two** addresses;
 - a **large organization** may be given **thousands** of addresses.
 - An **ISP**, as the Internet service provider, may be given **thousands or hundreds of thousands** based on the number of customers it may serve.

Restriction

- To simplify the handling of addresses, the Internet authorities impose **three restrictions** on classless address blocks:
 1. The addresses in a block must be contiguous, one after another.
 2. The number of addresses in a block must be a power of 2 (1, 2, 4, 8).
 3. The first address must be evenly divisible by the number of addresses.

Restriction



Contd.

- The above figure shows a block of addresses, in both **binary** and **dotted-decimal** notation, granted to a small business that needs **16 addresses**.
- We can see that the restrictions are applied to this block.
 - The addresses are **contiguous**.
 - The number of addresses is a power of 2 (**$16 = 2^4$**)
 - the first address is divisible by 16. The first address, when converted to a decimal number, is 3,440,387,360, which when divided by 16 results in 215,024,210.

Classless addressing Mask

- A better way to define a block of addresses is to select any address in the block and the mask.
- As we discussed before, a mask is a 32-bit number in which the n leftmost bits are 1s and the $32 - n$ rightmost bits are 0s.
- in classless addressing the mask for a block can take any value from 0 to 32. It is very convenient to give just the value of n preceded by a slash (***CIDR – Classless Inter Domain Routing notation***).

Classless addressing Mask

- The address and the $/n$ notation completely define the whole block (the first address, the last address, and the number of addresses).
- In IPv4 addressing, a block of addresses can be defined as $x.y.z.t/n$ in which $x.y.z.t$ defines one of the addresses and the $/n$ defines the mask.

Classless addressing Mask

- **First Address:**

- The first address in the block can be found by setting the $32-n$ right most bits in the binary notation of the address to 0s.

- **The last address in the block**

- can be found by setting the $32 - n$ rightmost bits in the binary notation of the address to 1s.

- **The number of addresses in the block**

- is the difference between the last and first address.
- It can easily be found using the formula 2^{32-n} .

2 Power Table

2^1	2
2^2	4
2^3	8
2^4	16
2^5	32
2^6	64
2^7	128

2^8	256
2^9	512
2^{10}	1024
2^{11}	2048
2^{12}	4096
2^{13}	8192
2^{14}	16384

Example 5:

- A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28.
 - What is the first address in the block?
 - What is the last address in the block?
 - The number of addresses in the block in the block?

Solution-Option 1

■ First Address

- The binary representation of the given address is :

11001101 00010000 00100101 00100111

- If we set 32–28 rightmost bits to 0, we get :

11001101 00010000 00100101 00100000

Or 205.16.37.32.

Solution-Option 1

- **Last Address :**

- The binary representation of the given address is

11001101 00010000 00100101 00100111

- If we set 32 – 28 rightmost bits to 1, we get

11001101 00010000 00100101 0010**1111**

or

205.16.37.47

Solution-Option 1

- **Number of Addresses :**
 - The value of n is 28
 - Number of addresses is 2^{32-28} or **16**.

Option Two

- Another way to find the first address, the last address, and the number of addresses is to represent the mask as a 32-bit binary (or 8-digit hexadecimal) number.
- This is particularly useful when we are writing a program to find these pieces of information. In the above example the /28 can be represented as

11111111 11111111 11111111 11110000

(twenty-eight 1s and four 0s).

Cont'd

Address : 11001101 00010000 00100101 00100111

Mask : 11111111 11111111 11111111 11110000

Find

- a. The first address
- b. The last address
- c. The number of addresses.

Solution-Option 2

A. the first address :

- The first address can be found by **ANDing** the given addresses with the mask.
- ANDing here is done bit by bit.

Address:	11001101	00010000	00100101	00100111
Mask:	11111111	11111111	11111111	11110000
First address:	11001101	00010000	00100101	00100000

Solution-Option 2

B. The last address:

- can be found by **ORing** the given addresses with the complement of the mask.

Address:	11001101	00010000	00100101	00100111
Mask complement:	00000000	00000000	00000000	00001111
Last address:	11001101	00010000	00100101	00101111

Solution-Option 2

C. The number of addresses:

- can be found by complementing the mask, interpreting it as a decimal number, and adding 1 to it.

Mask complement:	00000000 00000000 00000000 00001111
Number of addresses:	$15 + 1 = 16$

Subnetting

- During the era of **class full addressing**, subnetting was introduced.
- If an organization was granted a **large block in class A or B**, it could divide the addresses into several contiguous groups and assign each group to smaller networks (called **subnets**) or, in rare cases, share part of the addresses with neighbors.
- **Subnetting increases the number of 1s in the mask.**

Attributes of Subnetting

Network ID

Broadcast IP

First host IP

Last host IP

Next network

#IP addresses
in each block

CIDR/subnet

Network Address and Host Address

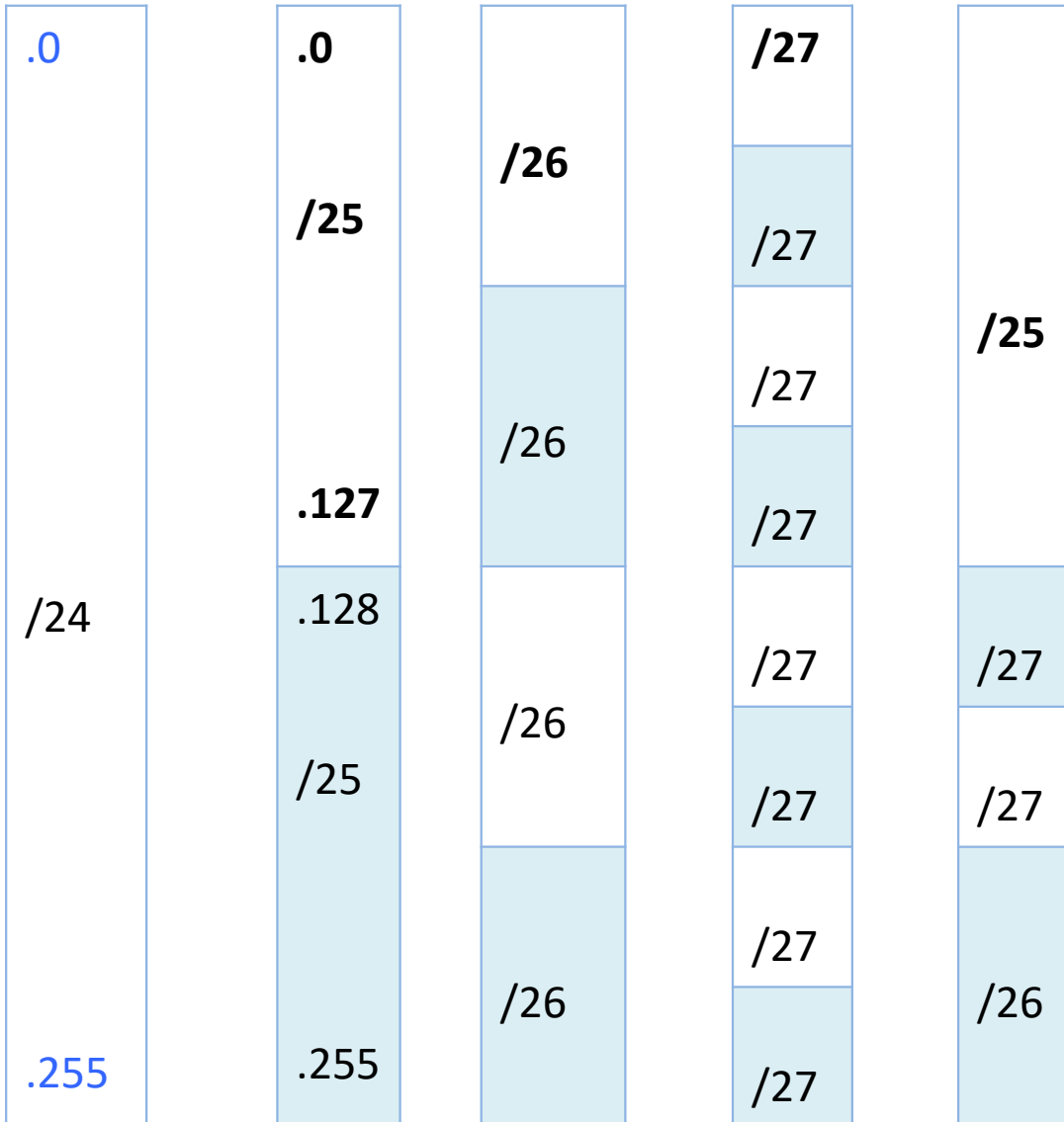
- When an organization is given a block of addresses, the organization is **free to allocate the addresses** to the devices that need to be connected to the Internet.
- It defines the organization itself to the **rest of the world**.
- The first address is the one that is used by **routers** to direct the message sent to the organization from the outside.

Network Address and Host Address

- The **Network address identifies the specific network to which host is attached**, and Host address uniquely identifies a host within a network.
- The distinction is important because router routes the packet according to a network address and don't care about host address.

Subnetting Types

FLSM (Fixed Length Subnet Masks) Subnetting	VLSM (Variable Length Subnet Masks) Subnetting
All subnets are equal in size.	Subnets are variable in size.
All subnets have equal number of hosts.	Subnets have variable number of hosts.
All subnets use same subnet mask.	Subnets use different subnet masks.
It is easy in configuration and administration.	It is complex in configuration and administration.
It wastes a lot of IP addresses.	It wastes minimum IP addresses.
It is also known as classfull Subnetting.	It is also known as classless Subnetting.
It supports both classfull and classless routing protocols.	It supports only classless routing protocols.



Subnetting table

Subnet	1	2	4	8	16	32	64	128	256
Host	256	128	64	32	16	8	4	2	1
CIDR	/24	/25	/26	/27	/28	/29	/30	/31	/32
Subnet Mask	0	128	192	224	240	248	252	254	255
3 rd octate	/16	/17	/18	/19	/20	/21	/22	/23	/24

Example:

Given: 192.168.4.0/24

Required : 3 subnets

Cont'd

Solution

Original networkID:
192.168.4.0/24

Subnet	1	2	4	8	16	32	64	128	256
Host	256	128	64	32	16	8	4	2	1
Subnet Mask	/24	/25	/26	/27	/28	/29	/30	/31	/32

Network ID	Subnet Mask	Host ID Range	# of Usable Host	Broadcast ID
192.168.4.0	/26	192.168.4.1-192.168.4.62	62	192.168.4.63
192.168.4.64	/26	192.168.4.65-192.168.4.126	62	192.168.4.127
192.168.4.128	/26	192.168.4.129-192.168.4.190	62	192.168.4.191
192.168.4.192	/26	192.168.4.193-192.168.4.254	62	192.168.4.255

Exercise

find all subnet attributes

10.1.1.55/28

Network ID:	10.1.1.48
Broadcast IP:	10.1.1.63
First host IP :	10.1.1.49
Last host IP :	10.1.1.62
Next network :	10.1.1.64
#IP addresses in each block :	16

Exercise

find all subnet attributes

10.1.1.37/29

Network ID:	10.1.1.32
Broadcast IP:	10.1.1.39
First host IP :	10.1.1.33
Last host IP :	10.1.1.38
Next network :	10.1.1.40
#IP addresses in each block :	8
	/29 255.255.255.248

Exercise

find all subnet attributes

10.2.2.20/30

Network ID:	10.2.2.20
Broadcast IP:	10.2.2.23
First host IP :	10.2.2.21
Last host IP :	10.2.2.22
Next network :	110.2.2.24
#IP addresses in each block :	4
	/27 255 255 255 252

Exercise

find all subnet attributes

10.2.2.199/26

10.2.2.0

10.2.2.64

10.2.2.128

10.2.2.192

10.2.3.0

10.2.3.64

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Exercise

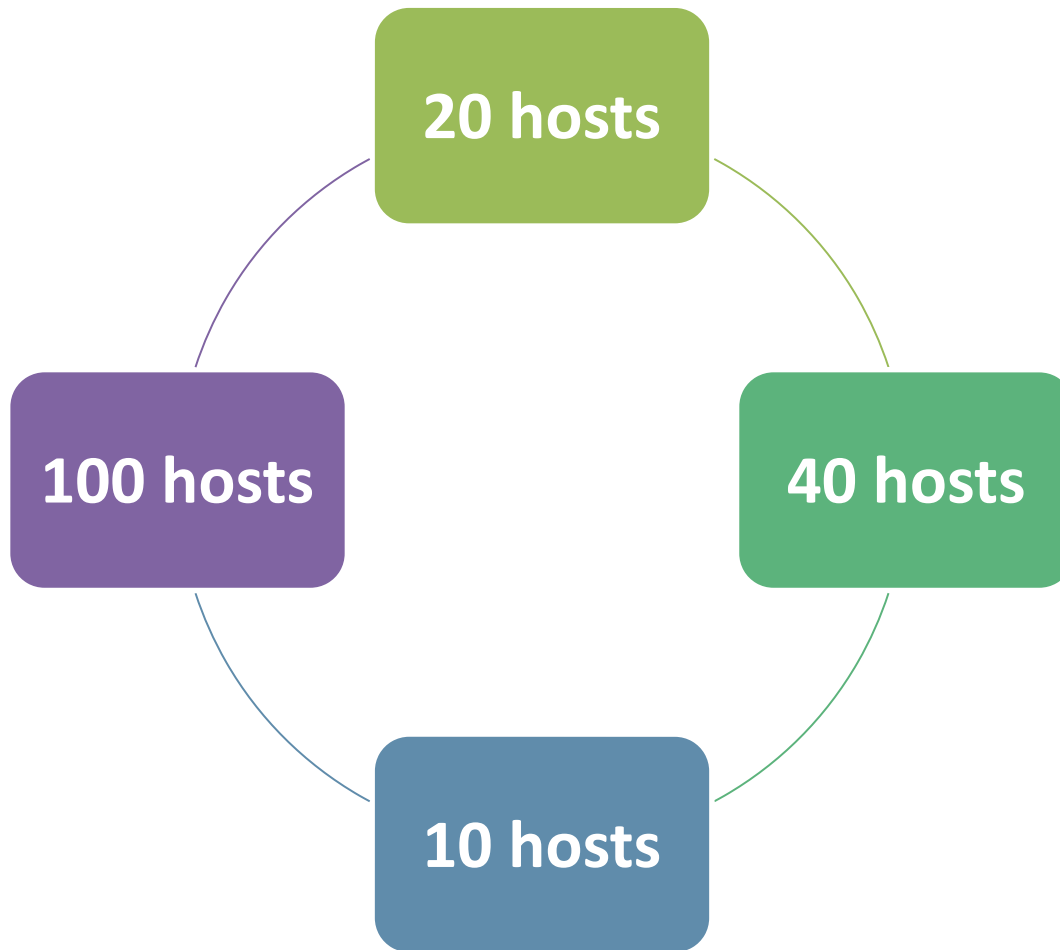
find all subnet attributes

10.2.2.199/26

Network ID:	10.2.2.192
Broadcast IP:	10.2.2.255
First host IP :	10.2.2.193
Last host IP :	10.2.2.254
Next network :	10.2.3.0
#IP addresses in each block :	64
	/26 255 255 255 192

Example

192.168.1.0



VLSM:100 hosts

- $2^h - 2 \geq 100$
 - $h=7$
- **\value**
 - $8+8+8+(8-7)=25$
- **Subnet mask**
 - 11111111.11111111.11111111.10000000
 - 255.255.255.128
- **Block size**
 - $2^7=128$
 - Range: 192.168.1.0/25 -192.168.1.127/25

VLSM: 40 hosts

- $2^h - 2 \geq 40$
 - $h=6$
- **\value**
 - $8+8+8+(8-6)=26$
- **Subnet mask**
 - 11111111.11111111.11111111.11000000
 - 255.255.255.192
- **Block size**
 - $2^6=64$
 - Range: 192.168.1.128/26 -192.168.1.191/26

VLSM : 20 hosts

- $2^h - 2 \geq 20$
 - $H=5$
- **\value**
 - $8+8+8+(8-5)=27$
- **Subnet mask**
 - 11111111.11111111.11111111.11100000
 - 255.255.255.224
- **Block size**
 - $2^5=32$
 - Range: 192.168.1.192/27 -192.168.1.223/27

VLSM: 10 hosts

- $2^h - 2 \geq 10$
 - $H=4$
- **\value**
 - $8+8+8+(8-4)=28$
- **Subnet mask**
 - 11111111.11111111.11111111.11110000
 - 255.255.255.240
- **Block size**
 - $2^4=16$
 - Range: 192.168.1.224/28 - 192.168.1.239/28

VLSM

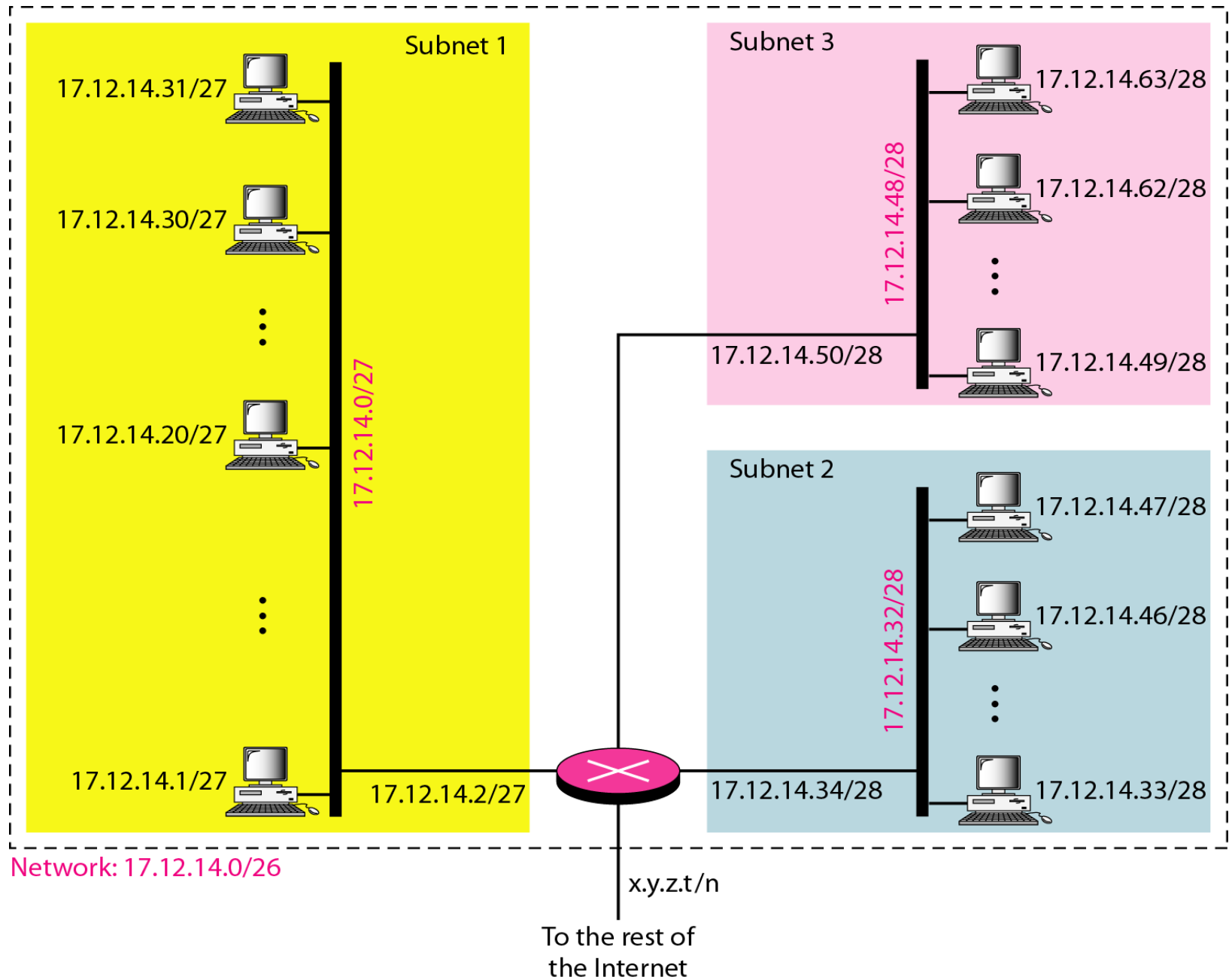
- 192.168.1.0/25 -192.168.1.127/25 for 100 hosts
- 192.168.1.128/26 -192.168.1.191/26 for 40 hosts
- 192.168.1.192/27 -192.168.1.223/27 for 20 hosts
- 192.168.1.224/28 -192.168.1.239/28 for 10 hosts

Example 6: Problem

- Suppose an organization is given the block 17.12.14.0/26, which contains 64 addresses. The organization has three offices and needs to divide the addresses into three sub blocks of 32, 16, and 16 addresses.

Example 6: Solution

- Suppose the mask for the first subnet is $n1$, then 2^{32-n1} must be 32, which means that $n1 = 27$.
- Suppose the mask for the second subnet is $n2$, then 2^{32-n2} must be 16, which means that $n2 = 28$.
- Suppose the mask for the third subnet is $n3$, then 2^{32-n3} must be 16, which means that $n3 = 28$.
- This means that we have the masks 27, 28, 28 with the organization mask being 26.



Address Allocation

- The next issue in classless addressing is address allocation.
- How are the blocks allocated?
- The ultimate responsibility of address allocation is given to a **global authority** called the **Internet Corporation for Assigned Names and Addresses (ICANN)**.
- However, ICANN does not normally allocate addresses to **individual organizations**. It assigns a large block of addresses to an ISP.

Address Allocation

- Each ISP, in turn, divides its assigned block into smaller sub blocks and grants the sub blocks to its customers.
- In other words, an ISP receives one large block to be distributed to its **Internet users**. This is called **address aggregation**.
- many blocks of addresses are aggregated in one block and granted to one ISP.

Example 7

- An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses).
- The ISP needs to distribute these addresses to three groups of customers as follows:
 - a. The 1st group has 64 customers; each needs 256 addresses.
 - b. The 2nd group has 128 customers; each needs 128 addresses.
 - c. The 3rd group has 128 customers; each needs 64 addresses.
- Design the sub blocks and find out how many addresses are still available after these allocations.

solution

Group 1

- For this group, each customer needs 256 addresses.
- This means that 8 bits are needed to define each host.
- The prefix length is then $32 - 8 = 24$. The addresses are

<i>1st Customer:</i>	<i>190.100.0.0/24</i>	<i>190.100.0.255/24</i>
<i>2nd Customer:</i>	<i>190.100.1.0/24</i>	<i>190.100.1.255/24</i>
<i>...</i>		
<i>64th Customer:</i>	<i>190.100.63.0/24</i>	<i>190.100.63.255/24</i>
<i>Total = $64 \times 256 = 16,384$</i>		

solution

Group 2

- For this group, each customer needs 128 addresses.
- This means that 7 bits are needed to define each host.
- The prefix length is then $32 - 7 = 25$. The addresses are

<i>1st Customer:</i>	<i>190.100.64.0/25</i>	<i>190.100.64.127/25</i>
<i>2nd Customer:</i>	<i>190.100.64.128/25</i>	<i>190.100.64.255/25</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.127.128/25</i>	<i>190.100.127.255/25</i>
<i>Total = $128 \times 128 = 16,384$</i>		

Solution

Group 3

- For this group, each customer needs 64 addresses. This means that 6 bits are needed to each host.
- The prefix length is then $32 - 6 = 26$. The addresses are

<i>1st Customer:</i>	<i>190.100.128.0/26</i>	<i>190.100.128.63/26</i>
<i>2nd Customer:</i>	<i>190.100.128.64/26</i>	<i>190.100.128.127/26</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.159.192/26</i>	<i>190.100.159.255/26</i>
<i>Total = $128 \times 64 = 8192$</i>		

Number of granted addresses to the ISP: 65,536

Number of allocated addresses by the ISP: 40,960

Number of available addresses: 24,576

Reading Assignment:



**What is
NAT**

**How NAT
works**

**Why NAT is
important**

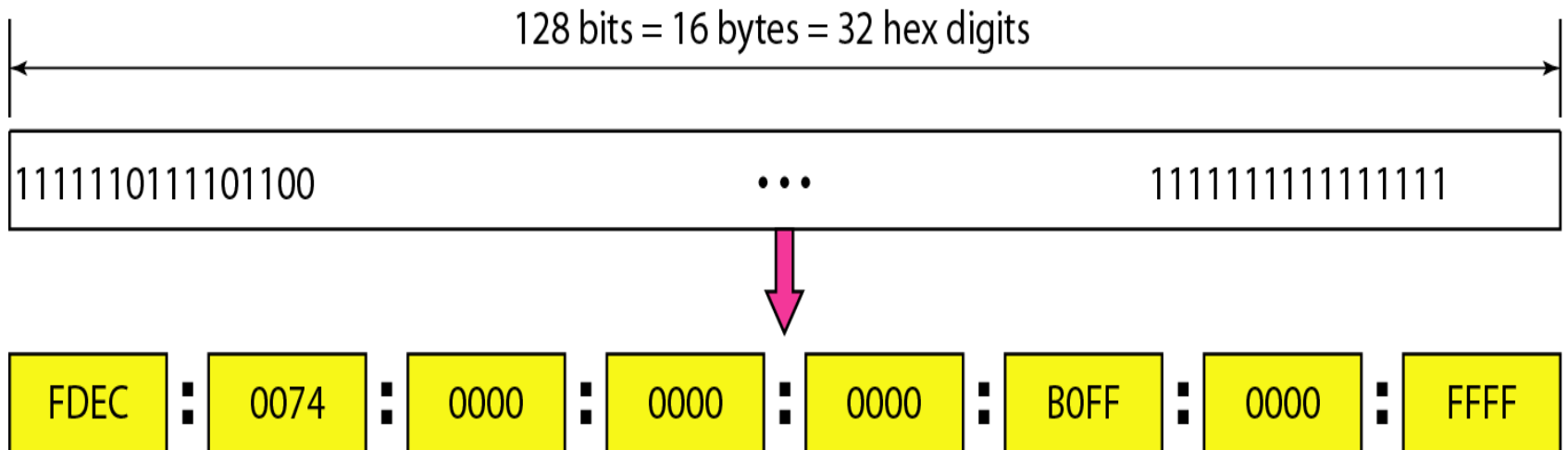
IPv6 ADDRESSES

- Despite all short-term solutions, address depletion is still a long-term problem for the Internet.
- This and other problems in the IP protocol itself have been the motivation for IPv6.
- An IPv6 address is 128 bits or 32 hexadecimal digits long.

Internet Protocol

IPV4	IPV6
32 bit	128 bit
4 octate	8 blocks
Decimal Representation	Hexadecimal Representation
1 ocate -8 bit	1 block -16bit
$4 * 8 = 32\text{bit}$	$8 * 16 = 128\text{bit}$
Subnet mask	No subnet mask
5 classes	No class
Use broadcast address	Not use any broadcast
Contain 0-255 devices	Contain 65,536 (FFFF)
Security add on	Embedded Security

IPv6 ADDRESSES



Abbreviated IPv6 addresses

Original

FDEC : 0074 : 0000 : 0000 : 0000 : B0FF : 0000 : FFF0



Abbreviated

FDEC : 74 : 0 : 0 : 0 : B0FF : 0 : FFF0



More abbreviated

FDEC : 74 : : B0FF : 0 : FFF0

Gap

Example 9

- Expand the address 0:15::1:12:1213 to its original.

Solution

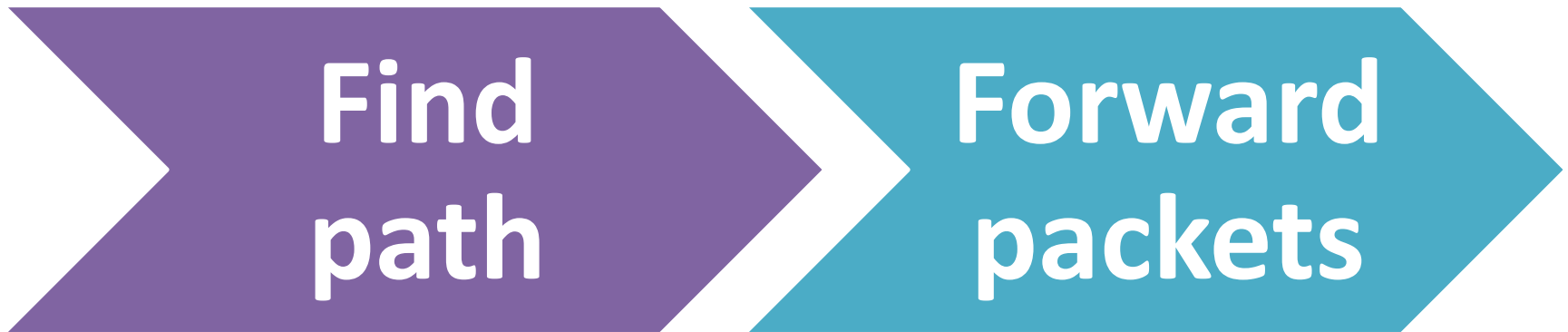
- We first need to align the left side of the double colon to the left of the original pattern and the right side of the double colon to the right of the original pattern to find how many 0s we need to replace the double colon.

XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX
0: 15: : 1: 12:1213

This means that the original address is.

0000:0015:0000:0000:0000:0001:0012:1213

What does a router do?



Routing vs Forwarding

- **Routing :**
 - Building maps and giving directions
- **Forwarding:**
 - Moving packets between interfaces according to the directions

IP Routing

- Path derived from information received from a routing protocol
- Several alternative paths may exist
 - best path stored in forwarding table
- Decisions are updated periodically or as topology changes (event driven)
- Decisions are based on:
 - topology
 - policies and metrics (hop count, filtering, delay, bandwidth, etc.)

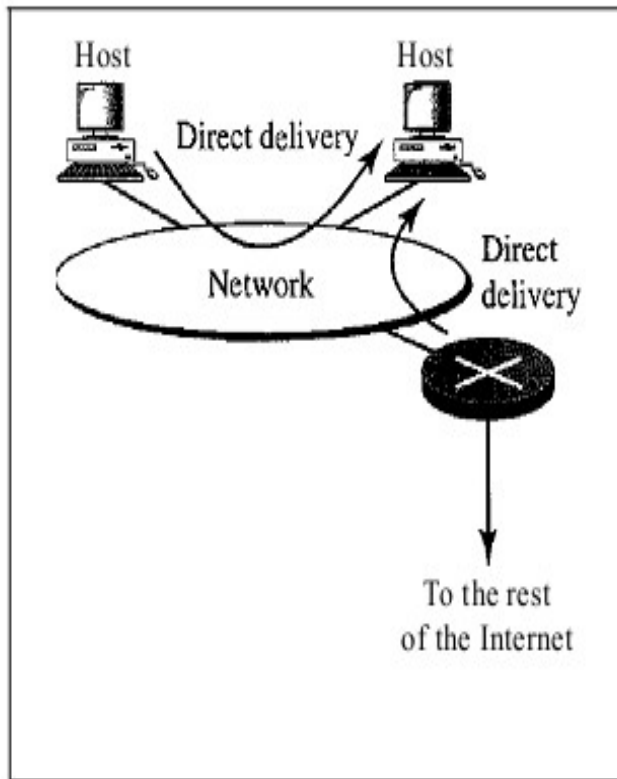
IP Forwarding

- Router decides which interface a packet is sent to
- Forwarding table populated by routing process
- Forwarding decisions:
 - destination address
 - class of service (fair queuing, precedence, others)
 - local requirements (packet filtering)
- Forwarding is usually aided by special hardware

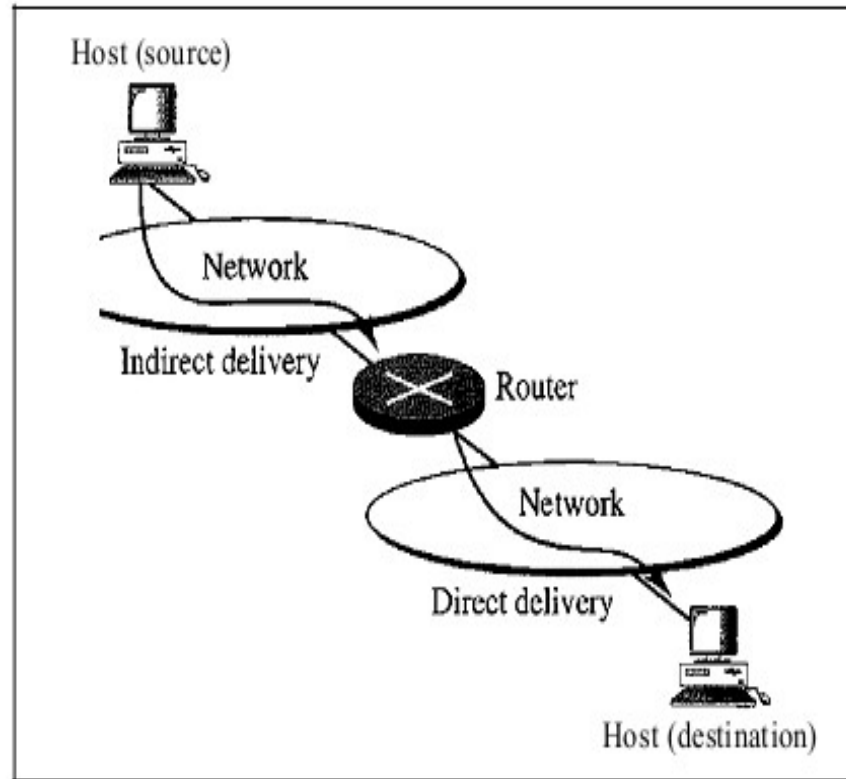
Contd.

- The delivery of a packet to its final destination is accomplished by using two different methods of delivery, **direct** and **indirect**.
- Direct delivery occurs when the source and destination of the packet are located on the same physical network or when the delivery is between the last router and the destination host

Contd.



a. Direct delivery



b. Indirect and direct delivery

Routing Table

- A host/router has routing table with an entry for each destination, or combination of destinations, to route IP packets.
- The routing table can be either **static** or **dynamic**.
- Depending on network requirement we can use :
 - **Static routing**
 - **Dynamic routing**
 - **a combination of both dynamic and static routing.**

Static Routing Table

- Contains information entered manually.
- Administrator enters the route for each destination into table.
- When a table is created, it **cannot update automatically** when there is a change in the Internet and the table must be manually altered by the administrator.
- Can be used in small internet that doesn't change very often, or in an experimental internet for troubleshooting.
- It is poor strategy to use a static routing table in a big internet such as the Internet.

Dynamic Routing Table

- Updated periodically by using one of the dynamic routing protocols such as RIP, OSPF, or BGP.
- Whenever there is a change in the Internet, such as a shutdown of a router or breaking of a link, the dynamic routing protocols update all the tables in the routers (and eventually in the host) automatically.
- The routers in a big internet such as the Internet need to be updated dynamically for efficient delivery of the IP packets.

Routing Table Format

- A routing table for classless addressing has a minimum of four columns.

Mask (In)	Network address	Next-hop address	Interface

- However, some of today's routers have even more columns.
- We should be aware that the number of columns is vendor-dependent, and not all columns can be found in all routers.

Mask	Network address	Next-hop address	Interface		Reference count	Use

Routing Table Format

- **Mask:-** *defines the mask applied for the entry.*
- **Network address:-** *defines network address to which packet is finally delivered.*
- **Next-hop address:-** *defines address of next-hop to which packet is delivered*
- **Interface:-** *shows the name of the interface.*
- **Flags:-** *on/off switches that signify either presence/absence. There 5 flags*
 - U (up):- indicates the router is up and running*
 - G (gateway):- means that the destination is in another network*
 - H (host-specific):- indicates that the entry in the network address field is a host-specific address.*
 - D (added by redirection):- indicates that routing information for this destination has been added to the host routing table by a redirection message from ICMP.*
 - M (modified by redirection):- indicates that the routing information for this destination has been modified by a redirection message from ICMP.*
- **Reference count:-** *gives the number of users of this route at the moment.*
- **Use:-** *shows the number of packets transmitted through this router*

Routing Protocols

- Routing protocols have been created in response to the demand for dynamic routing tables.
- A routing protocol is a combination of rules and procedures that lets routers in the internet inform each other of changes.
- It allows routers to share whatever they know about the internet or their neighborhood.

Routing Protocols

- Routing protocols insure that packet takes the best path from these paths.
- To accomplish this task, a router will need to perform following:
 - Learn about the directly connected routers on its interface
 - Find the networks that they have
 - Build routing table from this information
 - Choose best network from routing table to send the packet

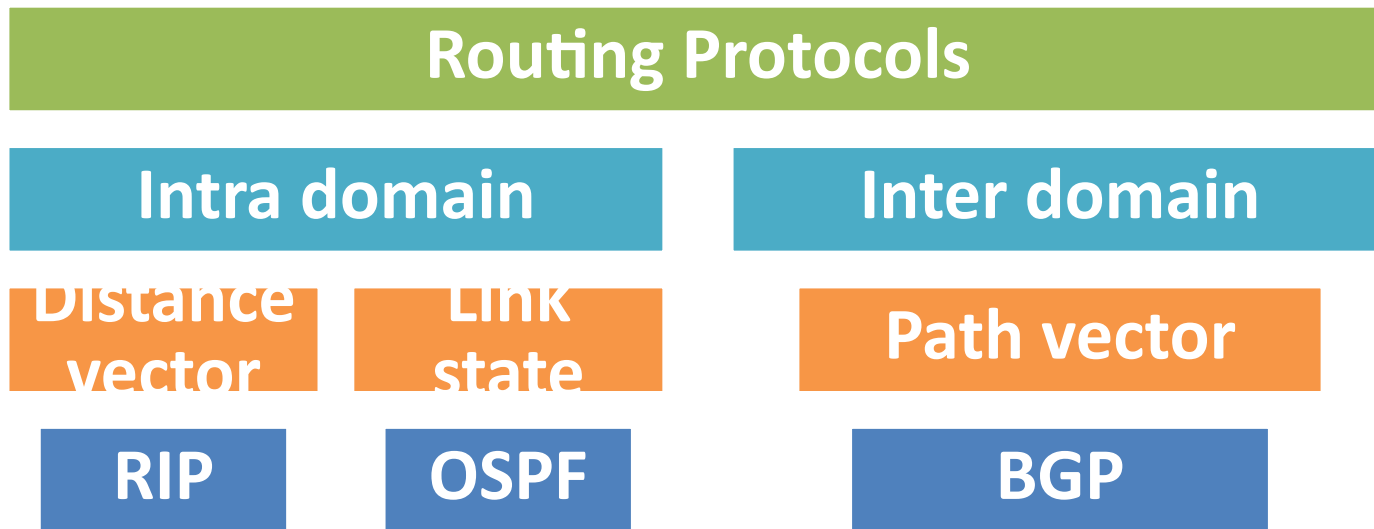
Purpose of Dynamic Routing Protocols

- Routing Protocols are used to facilitate the exchange of routing information between routers.
- The purpose of dynamic routing protocols includes:
 - Discovery of remote networks
 - Maintaining up-to-date routing information
 - Choosing the best path to destination networks
 - Ability to find a new best path if the current path is no longer available

Routing Updates

- Routing update is a mechanism of sharing information with neighboring routers.
- In a particular time duration router advertise its routing information through broadcast or multicast.
- Different protocols have different time intervals.
- Routing updates contain all necessary information for routing protocol such as learned network, timers, AS, AD, matrix values, interface details etc.

Types of Routing Protocols



Intra and Inter domain Routing

- Today, an internet can be so large that one routing protocol cannot handle task of updating the routing tables of all routers.
- For this reason, an internet is divided into autonomous systems (AS)

Autonomous systems (AS)

- AS is a group of networks and routers under the authority of a single administration.
- AS is a boundary line for routing protocol.
- It could be your company, or group of companies.
- It is defined by a numeric value.
- To distinguish between different AS, **Internet Assigned Numbers Authority (IANA)** provides a range from **1 to 65535**.

Intra domain vs Inter domain

- Routing inside an autonomous system is referred to as **intra domain routing** (Interior gateway protocols (IGPs))
- routing between autonomous systems is referred to as **inter domain routing** (Exterior gateway protocols (EGPs)).
- Each AS can choose **one or more intra domain** routing protocols to handle routing inside the AS. However, **only one inter domain** routing protocol handles routing between autonomous systems

Intra domain vs Inter domain

- Some examples of IGPs are:
 - RIPv1
 - RIPv2
 - EIGRP
 - IGRP
 - OSPF
- Example of EGPs is Border Gateway Protocol (BGP).

Distance Vector Routing

- It uses distance (metric value) and direction (vector) to find the best path to destination network
- In distance vector routing, the least-cost route between any two nodes is the route with minimum distance.
- In this protocol, as the name implies, each node maintains a vector (table) of minimum distances to every node.
- Shortest path

Distance Vector Routing

- The table at each node also guides the packets to the desired node by showing the next stop in the route (next-hop routing).
- We can think of nodes as the cities in an area and the lines as the roads connecting them.
- A table can show a tourist the minimum distance between cities.

Cont....

To	Cost	Next
A	0	—
B	5	—
C	2	—
D	3	—
E	6	C

A's table

To	Cost	Next
A	5	—
B	0	—
C	4	—
D	8	A
E	3	—

B's table

To	Cost	Next
A	3	—
B	8	A
C	5	A
D	0	—
E	9	A

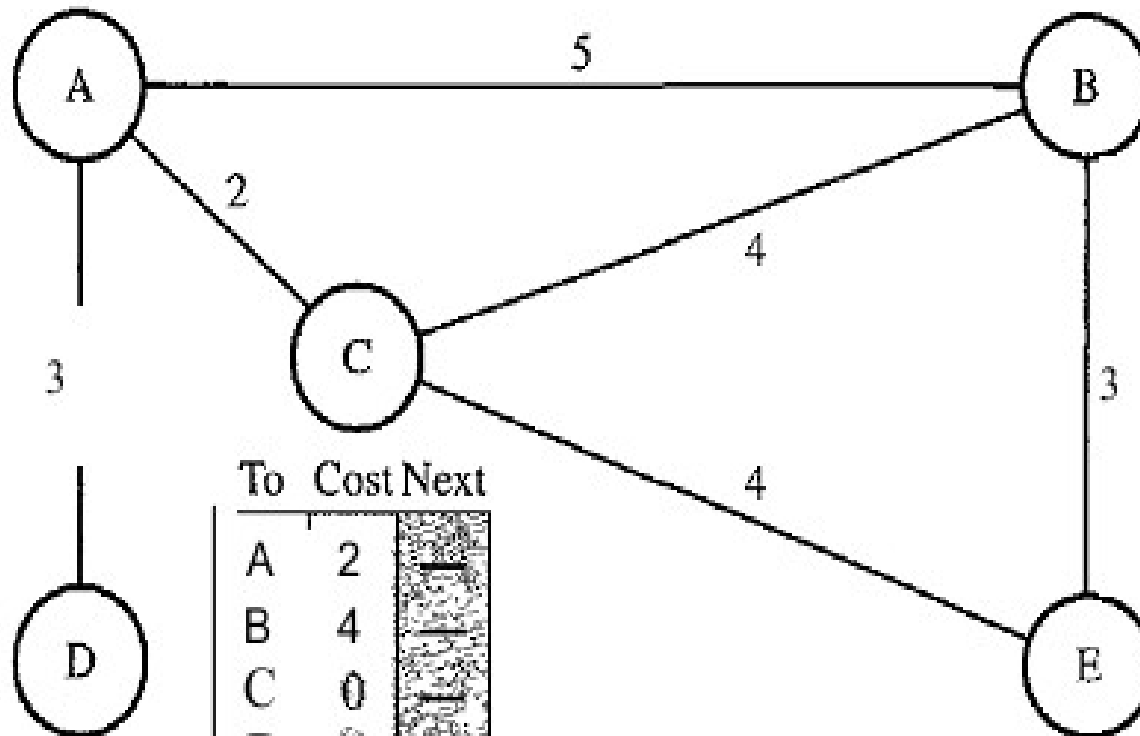
D's table

To	Cost	Next
A	2	—
B	4	—
C	0	—
D	5	A
E	4	—

e's table

To	Cost	Next
A	—	—
B	—	—
C	—	—
D	—	C
E	—	—

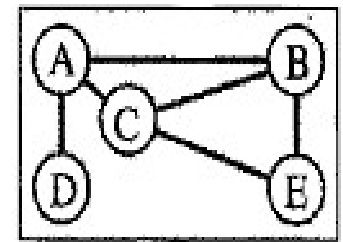
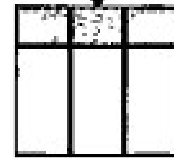
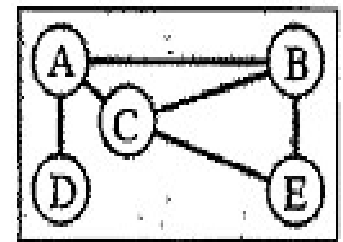
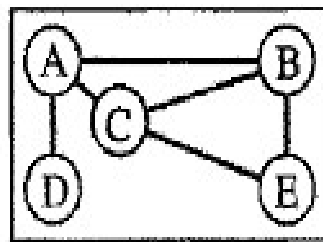
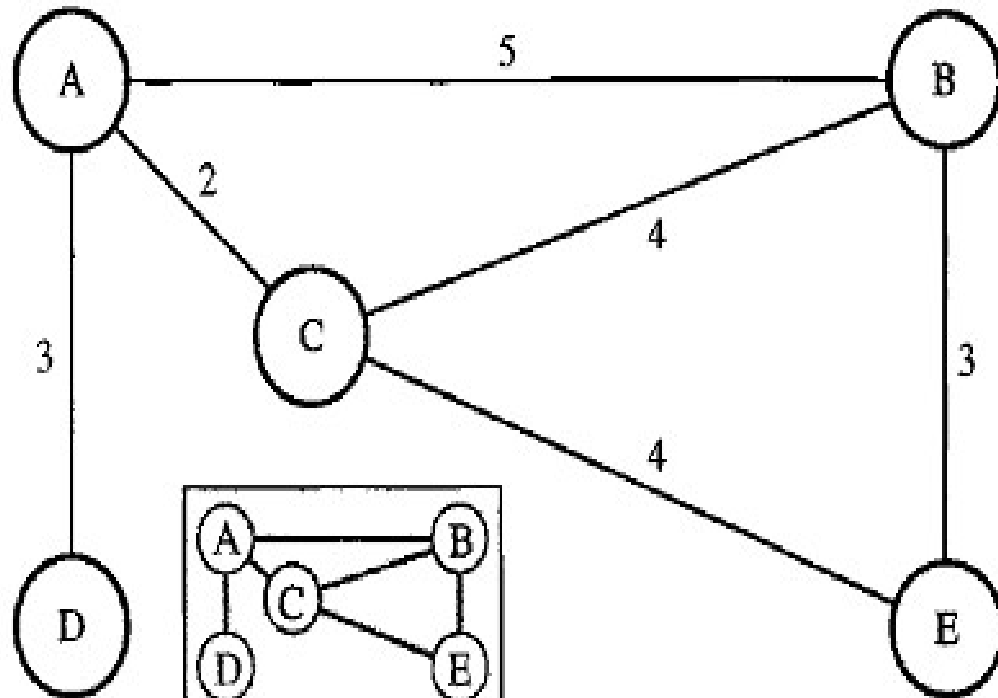
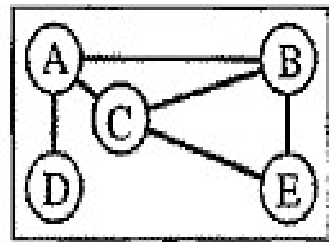
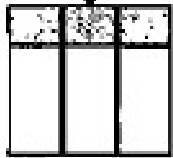
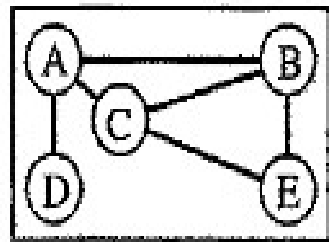
E's table



Link State Routing

- Link state routing protocols use more composite metric to locate the **best path** for destination network.
- It maintains three separate tables.
 - First table keeps track of directly connected neighbors.
 - Second table determines the entire network topology
 - the third is the routing table that keeps actual path.
- The node can use Dijkstra's algorithm to build a routing table
- **best path (consider the speed of the link)**
- OSPF is the example of link state protocol

Cont....



Distance vector and Link State

Distance Vector

- Routers communicate with **neighbor** routers advertising networks as measures of distance and vector
- Distance =metric ,
vector=direction(interface)
- Knowledge of the network from **directly connected neighbors**
- Send **periodic updates** of entire routing table
- Best for **simple, flat design ,non hierarchical networks ,minimum administration knowledge , convergence time is not an issue**

Link State

- Routers communicate with **all** other routers exchanging link state information to build a topology or the entire network
- Link-state=interface connections or “links” to other routers and networks
- Routers have **a complete view of the network** ,knowledge of the entire topology
- Send **triggered partial updates**
- Best for **large, hierarchical networks ,advanced administration knowledge ,convergence time is crucial**

Thank You